

SAMIDDHA JANA

West Bengal, India

📞 9382703483

✉ janasamiddha2006@gmail.com

🌐 LinkedIn: Jana Samiddha

Education

Midnapore College (Autonomous)

Bachelor of Science, Major in Physics

Expected 2027

Midnapore, India

- Currently in 3rd Year; SGPA: 8.35

Higher Secondary Education (Class XII)

Overall Score: 86% | Mathematics: 96

2024

Secondary Education (Class X)

Overall Score: 96% | Physical Science: 100, Mathematics: 100

2022

Research Interests

- Theoretical High Energy Physics & Astrophysics
- Quantum Computing & Information Theory
- Condensed Matter Physics (Theoretical)

Relevant Coursework

- Quantum Mechanics
- Classical Mechanics
- Special Relativity
- Tensor Analysis
- Mathematical Physics
- Linear Algebra
- Calculus
- Laser Physics

Experience

IIT Kharagpur – Translational NanoDevices Lab

Research Intern, under Dr. Shaili Sett, Centre for Interdisciplinary and Convergence Technology

June 2026

Kharagpur, India

- Characterized MoS₂-based FET phototransistors through extensive experimental and computational analysis.
- Fitted asymmetric I-V curves using a dual-transport model (2D thermionic emission with dual Schottky barriers and asymmetric Fowler-Nordheim tunneling), employing self-consistent damped fixed-point iteration for series resistance extraction.
- Performed biexponential photocurrent decay fitting with amplitude-weighted effective lifetime, and weighted nonlinear least-squares power-law fitting of photocurrent vs. optical power.
- Extracted publication-ready device parameters: ideality factor (η), Schottky barrier heights (ϕ_1 , ϕ_2), series resistance (R_S), specific detectivity (D^*), EQE, mobility, subthreshold swing, and I_{ON}/I_{OFF} ratio.

Workshops & Training

- **Online Workshop, QMet, IIT Bombay** – Completed a 2 hours online workshop on Quantum Technology, lectures are given by expert IIT faculties.
- **Winter School, IISc Bengaluru** – Completed a 5-day Winter School organized by the Department of Physics and passed the final evaluation.
- **Quantum Computing and Algorithms Workshop, IIT Kharagpur Research Park** – Hands-on training in quantum circuits, algorithms, and VQE for physical and molecular systems, with exposure to quantum hardware.
- **Photonics Research Laboratory Visit, IIT Kharagpur** – Gained knowledge of laser operation, photolithography, and photonic crystals.
- **Interdisciplinary Workshop on Neuroscience, Brain Mapping, and Cancer Biology** – Two-day workshop developing computational thinking and early biophysics research exposure through physics-based, quantitative analysis of neural systems.
- **Certificate Course, Haldia Government College** – Completed a course conducted by national and international faculty with an A+ grade.

Technical Skills

Languages: Python, SciLab

Typesetting: L^AT_EX

Other: Problem Solving, Analytical Thinking, Laboratory Instrument Handling, Team Collaboration

Achievements & Extracurriculars

Scholarships

NMMSE Scholarship Holder (2019–2023); Swami Vivekananda Scholarship Holder (2021–2023, 2024–Present)

2019 – Present

Competitive Examinations

Cleared JEE Mains 2024 and appeared for JEE Advanced 2024

2024

Hobbies

Philosophical Thinking, Reading Advanced Physics Texts, Watching Scientific Discussion, Chess, Outdoor Sports, Video Games